

SPANS-08: Cost-Efficient DQL Queries

Series: SPANS — Distributed Tracing and Spans | **Notebook:** 8 of 8 | **Created:** December 2025 | **Last Updated:** 04/25/2026

Optimizing Span Queries for Performance and Cost Efficiency

This notebook covers best practices for writing cost-efficient DQL queries, minimizing DDU consumption while maintaining effective observability.

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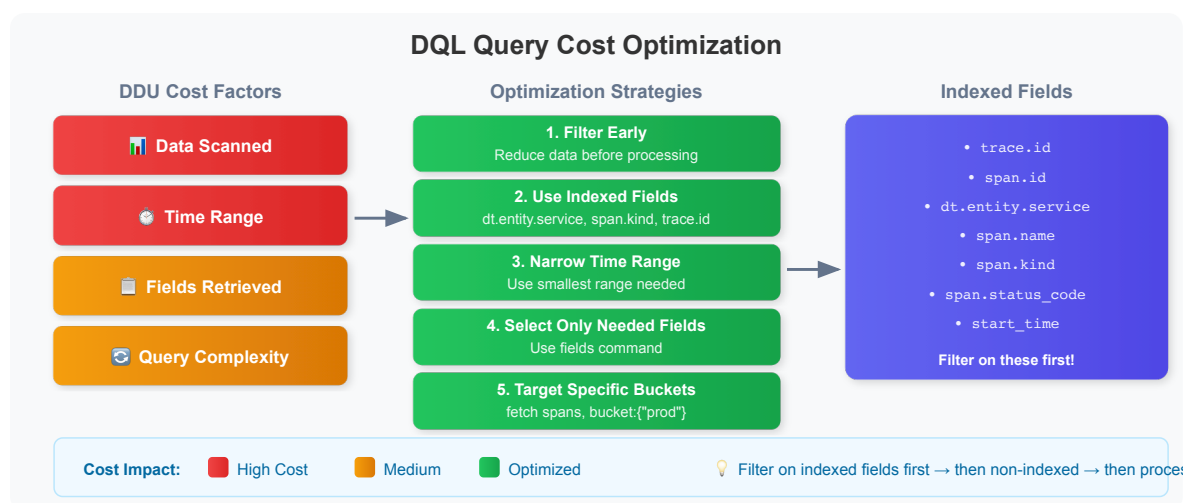
Prerequisites

Before starting this notebook, ensure you have:

-  Completed previous SPANS notebooks (01-07)
-  Understanding of DQL fundamentals
-  Access to span data in your tenant

1. Understanding Query Costs

DQL queries consume resources measured in DPS (Dynatrace Intelligence Processing Seconds) on current licensing, or DDUs (Dynatrace Intelligence Data Units) on classic contracts based on several factors:



Cost Optimization Priorities

Priority	Strategy	Impact
1	Filter early	High - reduces data scanned
2	Use indexed fields	High - leverages optimized indexes
3	Limit time range	High - reduces data volume
4	Select only needed fields	Medium - reduces transfer
5	Query specific buckets	Medium - targets data
6	Use aggregations	Low - summarizes results

2. Filter Early Pattern

The **most important** optimization: Apply filters as early as possible in the query pipeline.

```
// ✅ EFFICIENT: Filter BEFORE any processing
fetch spans, from:-1h
| filter span.kind == "server"           // Filter first!
| filter dt.entity.service == "SERVICE-6C36694E683AD694" // Further narrow
| filter span.status_code == "error"     // Even more specific
| fields start_time, span.name, duration
| limit 100
```

```
// ❌ INEFFICIENT: Processing before filtering scans more data
// This pattern wastes resources - shown for comparison only
fetch spans, from:-1h
| fieldsAdd duration_ms = duration / 1000000 // Processing first (bad)
| summarize {avg_duration = avg(duration_ms)}, by:{dt.entity.service,
span.name}
| filter dt.entity.service == "SERVICE-6C36694E683AD694" // Too late!
| limit 100
```

```
// ✅ EFFICIENT: Combine multiple filter conditions
fetch spans, from:-1h
| filter span.kind == "server"
      and isNotNull(dt.entity.service)
      and duration > 100ms
| fields start_time, span.name, duration
| sort duration desc
| limit 50
```

3. Indexed Fields for Performance

Filter on **indexed fields** for best query performance. These fields have optimized storage that allows fast filtering.

Commonly Indexed Span Fields

Indexed vs Non-Indexed Span Fields

Filter on indexed fields first for optimal query performance

✓ INDEXED (Fast Filtering)	
trace.id	Primary trace identifier
span.id	Individual span identifier
dt.entity.service	Service entity ID
span.name	Operation name
span.kind	server, client, internal...
span.status_code	ok, error

⚠ NON-INDEXED (Slower)	
url.path	Requires full scan
http.url	Use after indexed filters
db.statement	Custom attributes

Best Practice Pattern

```
fetch spans
| filter span.kind == "server"           ← indexed
| filter contains(url.path, "api")      ← then non-indexed
```

💡 **Tip:** When possible, filter on `dt.entity.service` rather than `service.name` - the entity ID is more reliably indexed.

```
// ✓ FAST: Filter on indexed fields
fetch spans, from:-1h
| filter isNotNull(dt.entity.service) // Indexed
| filter span.kind == "server"       // Indexed
| filter span.status_code == "error" // Indexed
| fields start_time, span.name, duration
| limit 100
```

```
// ⚠ SLOWER: Filter on non-indexed field requires full scan
fetch spans, from:-1h
| filter contains(url.path, "checkout") // Not indexed - slower
| limit 100
```

```
// ✓ BETTER: Combine indexed filter first, then non-indexed
fetch spans, from:-1h
| filter span.kind == "server" // Indexed - reduces data first
| filter isNotNull(dt.entity.service) // Indexed - further reduces
| filter contains(url.path, "checkout") // Non-indexed on smaller dataset
| limit 100
```

4. Field Selection

Select only the fields you need - avoid fetching all attributes.

```
// ✅ EFFICIENT: Select only required fields
fetch spans, from:-1h
| filter span.kind == "server"
| fields start_time,
          dt.entity.service,
          span.name,
          duration,
          span.status_code
| limit 100
```

```
// ❌ INEFFICIENT: Fetching all fields (default behavior)
// Returns ALL fields for each span - more data transfer
fetch spans, from:-1h
| filter isNotNull(dt.entity.service)
| limit 5
```

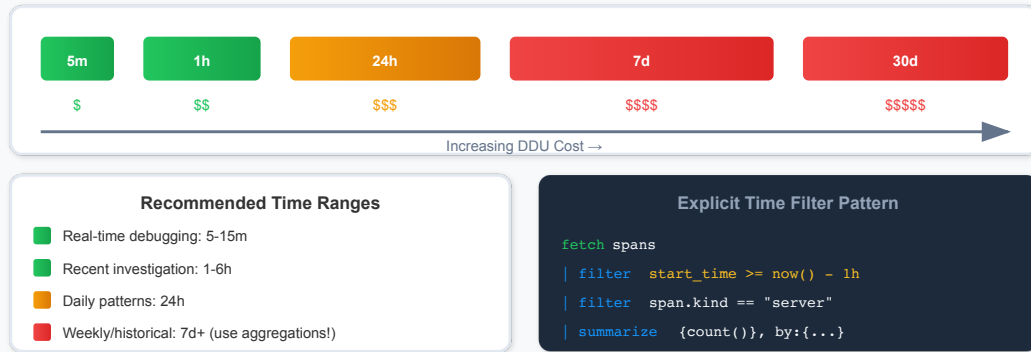
```
// ✅ EFFICIENT: Compute fields only when needed
fetch spans, from:-1h
| filter span.kind == "server"
| filter duration > 500ms
| fieldsAdd duration_ms = duration / 1000000
| fields start_time, dt.entity.service, span.name, duration_ms
| sort duration_ms desc
| limit 50
```

5. Time Range Optimization

Always use the smallest time range that meets your needs.

Time Range Optimization

Shorter time ranges = lower DDU cost | Use the minimum range needed



```
// Query with explicit time filter (last 15 minutes)
fetch spans, from:-1h
| filter start_time >= now() - 15m
| filter span.kind == "server"
| summarize {span_count = count()}, by:{dt.entity.service}
| sort span_count desc
| limit 20
```

```
// Narrow time range for troubleshooting specific incident
fetch spans, from:-1h
| filter start_time >= now() - 30m
| filter span.kind == "server"
| filter span.status_code == "error"
| fields start_time, dt.entity.service, span.name, span.status_message
| sort start_time desc
| limit 100
```

```
// Use aggregations for longer time ranges to reduce output
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
  span_count = count(),
  error_count = countIf(span.status_code == "error"),
  avg_duration_ms = avg(duration) / 1000000
}, by:{dt.entity.service}
| sort span_count desc
| limit 25
```

6. Aggregation Efficiency

Use aggregations to summarize data instead of retrieving raw records. Simpler aggregations are faster.

```
// ✅ EFFICIENT: Basic aggregations
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    requests = count(),
    errors = countIf(span.status_code == "error")
}, by:{dt.entity.service}
| sort requests desc
| limit 20
```

```
// ✅ EFFICIENT: Summarize at source with error rate calculation
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    total_requests = count(),
    error_count = countIf(span.status_code == "error"),
    p50_duration_ms = percentile(duration, 50) / 1000000,
    p95_duration_ms = percentile(duration, 95) / 1000000
}, by:{dt.entity.service}
| fieldsAdd error_rate_pct = (error_count * 100.0) / total_requests
| sort total_requests desc
| limit 20
```

```
// ⚠️ MORE EXPENSIVE: Many percentiles are costlier
// Use only the percentiles you actually need
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    p50 = percentile(duration, 50) / 1000000,
    p75 = percentile(duration, 75) / 1000000,
    p90 = percentile(duration, 90) / 1000000,
    p95 = percentile(duration, 95) / 1000000,
    p99 = percentile(duration, 99) / 1000000
}, by:{dt.entity.service}
| limit 20
```

```
// Time-bucketed aggregations for trends
fetch spans, from:-1h
| filter span.kind == "server"
| fieldsAdd time_bucket = bin(start_time, 5m)
| summarize {
    request_count = count(),
    avg_duration_ms = avg(duration) / 1000000
}, by:{time_bucket, dt.entity.service}
| sort time_bucket desc, request_count desc
| limit 100
```

7. High-Cardinality Grouping

Avoid grouping by high-cardinality fields (fields with many unique values).

```
// ❌ BAD: Grouping by high-cardinality field
// trace.id could have millions of unique values!
fetch spans, from:-1h
| summarize {trace_span_count = count()}, by:{trace.id}
| limit 10
```

```
// ✅ GOOD: Group by lower-cardinality fields
fetch spans, from:-1h
| summarize {span_count = count()}, by:{dt.entity.service, span.name}
| sort span_count desc
| limit 20
```

```
// ✅ GOOD: If you need trace analysis, filter first
fetch spans, from:-1h
| filter span.status_code == "error" // Reduce first
| filter start_time >= now() - 15m // Narrow time
| summarize {
    span_count = count(),
    services = collectDistinct(dt.entity.service)
}, by:{trace.id}
| sort span_count desc
| limit 20
```


8. Production Query Patterns

Optimized query templates for common production use cases.

```
// Production Pattern: Service Health Dashboard
// Optimized for dashboard refresh (low cost)
fetch spans, bucket: {"default_spans"}
| filter span.kind == "server"
| summarize {
    requests = count(),
    errors = countIf(span.status_code == "error"),
    avg_latency_ms = avg(duration) / 1000000
}, by:{dt.entity.service}
| fieldsAdd error_rate = (errors * 100.0) / requests
| sort requests desc
| limit 20
```

```
// Production Pattern: Error Investigation
// Fast targeted query for debugging
fetch spans, bucket: {"default_spans"}
| filter span.kind == "server"
| filter span.status_code == "error"
| filter start_time >= now() - 30m
| fields start_time,
    dt.entity.service,
    span.name,
    span.status_message,
    trace.id
| sort start_time desc
| limit 50
```

```
// Production Pattern: Latency Trend Analysis
// Using bin() for time-series without makeTimeseries
fetch spans, bucket: {"default_spans"}
| filter span.kind == "server"
| filter isNotNull(dt.entity.service)
| fieldsAdd time_bucket = bin(start_time, 5m)
| summarize {
    request_count = count(),
    p95_latency_ms = percentile(duration, 95) / 1000000
}, by:{time_bucket}
| sort time_bucket desc
| limit 50
```

```
// Production Pattern: Top Slow Operations
// Focus on actionable data
fetch spans, bucket: {"default_spans"}
| filter span.kind == "server"
| filter duration > 1s
| summarize {
    occurrence_count = count(),
    avg_duration_ms = avg(duration) / 1000000,
    max_duration_ms = max(duration) / 1000000
}, by:{dt.entity.service, span.name}
| sort avg_duration_ms desc
| limit 20
```

```
// Production Pattern: Well-optimized comprehensive query
// Follows all optimization rules
fetch spans, from:-1h
| filter start_time >= now() - 1h // 1. Appropriate time range
| filter isNotNull(dt.entity.service) // 2. Filter early (indexed)
| filter span.kind == "server" // 2. Additional early filter (indexed)
| filter span.status_code == "error" // 3. Indexed field
| fields start_time, span.name, duration, http.route // 4. Only needed fields
| summarize {
    error_count = count(),
    avg_duration_ms = avg(duration) / 1000000
}, by:{span.name, http.route} // 5. Low cardinality
| sort error_count desc
| limit 20 // 6. Limited results
```

9. Performance Checklist

Use this checklist before running production queries:

Query Optimization Checklist:

- ❑ 1. Time range – Is it as short as possible for my needs?
- ❑ 2. Filter early – Are filters immediately after fetch?
- ❑ 3. Indexed fields – Am I filtering on indexed fields first?
 - trace.id, span.id, dt.entity.service
 - span.name, span.kind, span.status_code
 - start_time
- ❑ 4. Field selection – Am I selecting only needed fields?
- ❑ 5. Low cardinality – Are my group-by fields low cardinality?
- ❑ 6. Result limits – Do I have appropriate limits?
- ❑ 7. Bucket targeting – Am I querying specific buckets if known?

Summary

In this notebook, you learned:

- ✅ **Query cost factors** and optimization priorities
- ✅ **Filter early pattern** to reduce data scanned
- ✅ **Indexed fields** for faster query execution
- ✅ **Field selection** to minimize data transfer
- ✅ **Time range optimization** for cost control
- ✅ **Aggregation efficiency** for summarized results
- ✅ **High-cardinality grouping** pitfalls to avoid
- ✅ **Production patterns** for common use cases
- ✅ **Performance checklist** for query review

Series Complete! 🎉

You have completed the **Spans & Distributed Tracing** notebook series!

What You've Learned:

1. **Fundamentals** - Span structure and distributed tracing concepts
2. **Querying** - DQL syntax for effective span analysis

3. **Troubleshooting** - Error detection and root cause analysis
4. **Topology** - Service dependencies and flow visualization
5. **Analytics** - Advanced metrics and trend analysis
6. **Security** - Security monitoring and compliance with spans
7. **Buckets & Pipeline** - Data architecture, OpenPipeline, and governance
8. **Cost Optimization** - Efficient query patterns and indexed fields

Next Steps:

- Apply these patterns to your own Dynatrace environment
- Build dashboards using the optimized query patterns
- Configure alerts based on span analytics
- Set up OpenPipeline for data optimization
- Explore Dynatrace Intelligence integration for intelligent analysis

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